

The 9-Tuple Formation of Constrained Object Hierarchies: The Mathematical Foundation and Implications for Artificial Intelligence

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Abstract

Research on artificial general intelligence (AGI) lacks a unified mathematical foundation that integrates hierarchical structure, adaptive learning, and constraint-regulated behavior; prevailing symbolic, neural, dynamical, and probabilistic paradigms remain fragmented. Constrained Object Hierarchies (COH) are introduced as a neuroscience-informed and mathematically rigorous formalism in which intelligence arises from hierarchical composition governed by adaptive constraints. Central to COH is a minimal 9-tuple that specifies components, semantics, constraints, and learning. This work develops the full mathematical foundation of the 9-tuple and proves its soundness, completeness, expressive minimality (in a functionally non-redundant sense), and orthogonality (as non-redundancy of components). The framework is situated within category-theoretic, dynamical-systems, and information-theoretic perspectives; it is shown to admit a categorical formulation, to instantiate a constrained hybrid dynamical system, and to serve both as a universal hierarchical approximator and as a constrained optimization architecture. Four AGI-relevant theorems provide quantitative guarantees on world-model fidelity, behavioral stability, generalization robustness, agentic decision complexity, and scalable hierarchical composition. Cross-domain instantiations spanning physical, biological, cyber-physical, economic, and cultural-computational systems demonstrate representational adequacy, theoretical coherence, and internal consistency. Together, these results establish the COH 9-tuple as a foundational unit for modeling intelligence and a unified, scalable, safety-aligned framework for AGI research, integrating hierarchical composition with adaptive optimization under constraints.

Keywords: *artificial general intelligence; hierarchical models; constraints; category theory; hybrid dynamical systems; universal approximation; optimization; safety and interpretability.*

1. Introduction

Artificial General Intelligence (AGI) research seeks a principled account of how intelligent behavior can emerge across diverse tasks, environments, and representational settings. Although significant advances have been made in symbolic reasoning, neural computation, reinforcement learning, cognitive architectures, and probabilistic inference, existing approaches remain fragmented. Each captures an

important facet of intelligence—structure, learning, prediction, or control—but none offers a unified mathematical framework capable of integrating these facets into a coherent theory of general intelligence.

What problem does COH solve? The central challenge is that existing frameworks excel in one dimension but fail to integrate others: neural networks learn but lack explicit hierarchical structure and formal safety guarantees; symbolic systems compose cleanly but struggle with adaptation and uncertainty; constrained optimization provides safety but without learning or composition. COH addresses this gap by providing a single formalism in which hierarchy, learning, and constraints coexist and interoperate.

Constrained Object Hierarchies (COH) (Wang, 2025) address this gap by grounding intelligence in three neuroscience-inspired principles: hierarchical organization, predictive optimization under uncertainty, and constraint-driven regulation, which together support both stability and adaptive behavior. Contemporary research in cortical hierarchy, predictive coding, and homeostatic regulation (Rao et al., 2024; Miyashita, 2024; Lin et al., 2025) supports the view that intelligent systems can be modeled as structured, adaptive hierarchies whose behavior emerges from interactions among components, goals, and constraints.

COH formalizes these principles through a functionally non-redundant 9-tuple that defines components, attributes, methods, neural learning mechanisms, goals, constraints, daemons, and semantic embeddings within a single mathematical structure. This paper develops the complete theoretical foundation of COH, establishes its internal consistency, and demonstrates its expressive power across multiple scientific domains. In addition, we extend the core framework with three novel theoretical contributions: (1) quantitative bounds on world model fidelity, (2) formal guarantees against jagged intelligence through constraint-aligned generalization, and (3) constructive proofs for building verifiably safe agentic systems with polynomial-time decision complexity.

Contributions

This work makes the following key contributions:

- **A complete mathematical formalization of Constrained Object Hierarchies (COH)** as a functionally non-redundant 9-tuple specifying hierarchical structure, semantic composition, constraint systems, adaptive optimization, and learning mechanisms.
- **A unified category-theoretic foundation**, demonstrating that COH supports products, coproducts, morphisms, functorial mappings, and limit constructions required for rigorous compositional reasoning.
- **A constrained hybrid dynamical-systems formulation**, establishing stability properties, admissible constraint regions, and connections to information-theoretic principles within a coherent dynamical interpretation.
- **A unified learning and adaptation theory**, showing that COH functions as both a universal hierarchical approximation architecture and a constraint-regulated optimization system capable of supporting reinforcement learning and hierarchical credit assignment.

- **Formal guarantees of soundness, completeness, expressive minimality (functional non-redundancy), and orthogonality (non-redundancy of components)**, proving that COH is internally consistent, irreducible, and suitable as a general mathematical basis for intelligence.
- **Three AGI-relevant theorems providing quantitative guarantees:** (a) bounds on world-model fidelity as a function of constraint expressiveness; (b) cross-hierarchical smoothness conditions that prevent jagged intelligence behaviors; and (c) constructive methods for building coherent agentic systems with bounded decision complexity and verifiable safety properties.
- **Cross-domain demonstrations across six heterogeneous scientific and engineering systems**, showing that COH supports stable modeling and theoretical coherence across physical, biological, cyber-physical, economic, and cultural-computational domains.
- **Integration with major theories of intelligence**, including hierarchical reinforcement learning, cognitive architectures, category-theoretic compositionality, and active inference, positioning COH as a unified and mathematically rigorous foundation for AGI.

Paper Organization

Section 2 reviews prior work across neuroscience, mathematical modeling, constrained optimization, and AGI research. Section 3 formalizes the COH 9-tuple. Section 4 develops its category-theoretic foundations, and Section 5 presents the dynamical-systems perspective. Section 6 introduces learning and adaptation mechanisms. Section 7 establishes soundness, completeness, and functional minimality. Section 8 applies COH to six scientific domains. Section 9 relates COH to existing theories of intelligence, Section 10 discusses implications for AGI development, and Section 11 concludes.

2. Literature Review

The pursuit of Artificial General Intelligence (AGI) necessitates a formal framework that integrates structural composition, adaptive learning, and constraint satisfaction. While existing paradigms offer valuable insights, they remain fragmented. This review synthesizes foundational work from cognitive neuroscience, mathematical frameworks for intelligence, and computational models of constrained systems, establishing the intellectual context and gap that Constrained Object Hierarchies (COH) theory aims to fill.

2.1 Neuroscience Foundations for Hierarchical Intelligence

A core inspiration for structurally grounded AGI comes from principles of neural organization. The hierarchical processing observed in the primate visual cortex provides a biological blueprint for compositional intelligence regulation (Rao et al., 2024; Miyashita, 2024; Lin et al., 2025). These three pillars—hierarchy, predictive optimization, and constraint satisfaction—form the neuroscientific cornerstone of the COH framework.

2.2 Mathematical and Theoretical Frameworks for Intelligence

Category theory provides a powerful language for describing compositional structures and relationships, relevant for defining COH morphisms (Fong and Spivak, 2019). However, these mathematical tools have rarely been integrated into a unified theory of general intelligence.

2.3 World Modeling and Representation Learning

The problem of learning accurate world models from data has seen significant advances in deep learning, with architectures like transformers demonstrating remarkable sequence modeling capabilities (Vaswani et al., 2017). However, current approaches lack formal guarantees on model fidelity and often fail to capture hierarchical structure explicitly. Research on conceptual spaces (Gärdenfors, 2014) and semantic embeddings provides foundations for the COH embedding component $\$E\$$, but typically without the constraint-driven semantics that COH provides for ensuring model-world alignment.

2.4 Generalization and Jagged Intelligence

The problem of robust generalization—where systems perform well on training distributions but fail on distributionally shifted tasks—has been identified as "jagged intelligence" in recent AI safety literature (Hendrycks et al., 2022). Theoretical work on out-of-distribution generalization (Arjovsky et al., 2019) and domain adaptation provides mathematical frameworks, but these typically lack the compositional, constraint-based approach that COH employs to ensure smooth capability generalization across task spaces.

2.5 Agentic Systems and Safe Autonomy

Research on autonomous agents spans cognitive architectures (Miyashita, 2024), hierarchical reinforcement learning (Pateria et al., 2024), and constrained decision-making (Altman, 1999). While these provide important components for agent design, they lack a unified mathematical foundation that simultaneously addresses perception, action, goals, and ethics within a single formalism with complexity guarantees. Recent work on AI safety emphasizes constraint-based approaches (Amodei et al., 2016), but typically without the hierarchical composition and formal guarantees that COH provides.

2.6 The Role of Constraints in Intelligent Systems

Classical AI models constraints through CSPs (Russell & Norvig, 2021), while constrained optimization frameworks such as CMDPs (Altman, 1999) introduce bounds on expected costs. Neuroscience suggests constraints are fundamental—from homeostatic limits to syntactic structures—supporting the COH view that constraints must be first-class citizens.

2.7 Synthesis and Gap Identification

Prior work provides essential but isolated components: neuroscientific principles of hierarchy and prediction, mathematical tools for composition and dynamics, computational models for learning and planning, and formalisms for constraint handling. The key gap is a *unified, mathematically rigorous framework* that cohesively integrates these elements. COH theory addresses this by proposing a functionally non-redundant 9-tuple formalism that explicitly defines hierarchical objects (C), their properties (A), behaviors (M), and learning capacities (N), all under the governance of a multi-tiered constraint system (I,

T, G) maintained by active processes (D), with a unifying semantic embedding (E). This bridges the expressiveness of symbolic systems, the adaptability of neural networks, and the principled governance of constraint-based reasoning, offering a foundational model for AGI.

3. Formal Definitions of COH Components

3.1 Intuitive Glossary of the 9-Tuple

Before presenting the formal definitions, we provide an intuitive guide to the nine components:

Component	Name	Intuitive Role
C	Components	The hierarchical parts or sub-objects that compose the system (e.g., sensors, actuators, subsystems)
A	Attributes	State variables that describe the system's current condition (e.g., position, temperature, beliefs)
M	Methods	Executable actions or transformations that change the system's state
N	Neural Components	Learnable, parameterized functions that enable adaptation and prediction
E	Embedding	A unified semantic representation mapping the entire object to a vector space
I	Identity Constraints	Invariant conditions that must always hold (e.g., physical laws, safety boundaries)
T	Trigger Constraints	Event-condition-action rules that specify responses to specific situations
G	Goal Constraints	Optimization objectives that guide behavior toward desired outcomes
D	Constraint Daemons	Continuous monitoring processes that enforce constraints in real time

Notation note: Throughout this paper, $\$T\$$ is used for trigger constraints (Section 3.8) and also appears in time domains (Section 3.9). Context disambiguates these uses; a notation table is provided in Supplementary Appendix A.

3.2 The 9-Tuple Formalism

Definition 3.1 (Constrained Object Hierarchy). A COH is defined as a 9-tuple:

$$\mathcal{O} = (C, A, M, N, E, I, T, G, D)$$

where each component has specific mathematical structure.

3.3 Component Formalization

Definition 3.2 (Components). Let **Obj** be the category of COH objects. Then:

$$C: \mathcal{O} \rightarrow \mathcal{P}(\mathbf{Obj})$$

where $\mathcal{P}$ denotes the power set, with the additional constraint that the component relation forms a directed acyclic graph (DAG).

Definition 3.3 (Attributes). Attributes are state variables:

$$A = \{a_i: \mathcal{O} \rightarrow \mathcal{V}_i\}_{i=1}^n$$

where each a_i maps the object to a value in domain $\mathcal{V}_i$, forming a measurable space $(\Omega, \mathcal{F})$.

Definition 3.4 (Methods). Methods are executable transformations:

$$M = \{m_j: \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S} \times \mathcal{R}\}_{j=1}^k$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and $\mathcal{R}$ is the reward (or outcome) space.

Definition 3.5 (Neural Components). Neural components are parameterized functions:

$$N = \{f_\theta: \mathcal{X} \rightarrow \mathcal{Y} \mid \theta \in \Theta\}$$

where each f_θ is a measurable function parameterized by $\theta \in \Theta$, the parameter space.

Definition 3.6 (Embedding Component). The embedding function:

$$E: \mathcal{O} \rightarrow \mathcal{H}$$

maps the entire object to a Hilbert space $\mathcal{H}$, providing a unified semantic representation.

Definition 3.7 (Identity Constraints). Identity constraints are invariant conditions:

$$I = \{\phi_i: \mathcal{S} \rightarrow \mathbb{B}\}_{i=1}^p$$

where each ϕ_i is a predicate that must evaluate to true for all reachable states.

Definition 3.8 (Trigger Constraints). Trigger constraints are event-condition-action rules:

$$T = \{(e_i, c_i, a_i) \mid e_i \in \mathcal{E}, c_i: \mathcal{S} \rightarrow \mathbb{B}, a_i \in \mathcal{A}\}$$

where $\mathcal{E}$ is an event space.

Definition 3.9 (Goal Constraints). Goal constraints are optimization objectives:

$$G = \{g_j: \mathcal{S} \times \mathcal{A} \times \mathcal{T} \rightarrow \mathbb{R}\}_{j=1}^q$$

where $\mathcal{T}$ is a time domain, and goals may be composed as:

$$G_{\text{total}} = \sum_{j=1}^q w_j g_j + \lambda R(\theta)$$

with regularization term $R(\theta)$.

Definition 3.10 (Constraint Daemons). Constraint daemons are continuous monitoring processes:

$$D = \{d_k: \mathcal{S}^T \rightarrow \mathcal{A}\}_{k=1}^r$$

where each d_k operates on the state trajectory.

4. Category-Theoretic Foundations

4.1 COH as a Category

Definition 4.1 (Category of COH Objects). Let **COH** be the category defined as follows:

- **Objects.** Objects are COH 9-tuples $\mathcal{O}$.
- **Morphisms.** Morphisms

$$f: \mathcal{O}_1 \rightarrow \mathcal{O}_2$$

are structure-preserving maps that respect hierarchical composition and all associated constraints.

Theorem 4.1 (Compositionality)

The category **COH** has the following categorical structures:

1. **Finite products:**

$$\mathcal{O}_1 \times \mathcal{O}_2$$

representing **parallel composition**.

2. **Finite coproducts:**

$$\mathcal{O}_1 + \mathcal{O}_2$$

representing **alternative composition**.

3. **Exponentials:**

$$\mathcal{O}_2^{\mathcal{O}_1}$$

representing **function objects** between COHs.

Proof sketch: The product COH is constructed by taking Cartesian products of each component set, with appropriate constraint conjunction. The coproduct takes disjoint unions with constraint disjunction. Exponentials are constructed from method spaces with appropriate currying. Complete proofs are provided in Supplementary Appendix B.

4.2 Hierarchical Structure as a Functor

Definition 4.2 (Component Functor). The component structure defines a functor:

$$F: \mathbf{COH} \rightarrow \mathbf{Graph}$$

where **Graph** is the category of directed graphs, mapping each COH to its component DAG.

Theorem 4.2 (Hierarchy Preservation)

For any morphism

$$f: \mathcal{O}_1 \rightarrow \mathcal{O}_2$$

in **COH**, there exists a graph homomorphism

$$F(f): F(\mathcal{O}_1) \rightarrow F(\mathcal{O}_2)$$

that preserves the hierarchical structure.

4.3 Constraint Systems as Limits

Definition 4.3 (Constraint Category)

For a COH $\mathcal{O}$, define the **constraint category** $\mathbf{Con}(\mathcal{O})$ as follows:

- **Objects.** Objects are states

$$s \in \mathcal{S}$$

satisfying all identity constraints I .

- **Morphisms.** Morphisms are state transitions generated by methods M that respect trigger constraints T .

Theorem 4.3 (Constraint Satisfaction as Limit). The set of admissible states forms a limit in the diagram formed by constraint predicates:

$$\mathcal{S}_{\text{adm}} = \lim (\{\phi_i^{-1}(\text{true})\}_{i=1}^p)$$

5. Dynamical Systems Perspective

5.1 COH as a Hybrid Dynamical System

Definition 5.1 (COH Dynamics). A COH defines a hybrid dynamical system:

$$\begin{aligned}\frac{ds}{dt} &= f(s, a, \theta) \text{ for continuous evolution} \\ s_{t+1} &= g(s_t, a_t, \theta) \text{ for discrete transitions}\end{aligned}$$

with switching conditions defined by trigger constraints T .

Theorem 5.1 (Existence and Uniqueness). Under Lipschitz continuity assumptions on f and g , and measurability of trigger conditions, the COH dynamics admit unique solutions for given initial conditions and control policies.

5.2 Stability Analysis

Definition 5.2 (Constraint-Admissible Region). Let:

$$\mathcal{R} = \{s \in \mathcal{S} \mid \forall \phi \in I, \phi(s) = \text{true}\}$$

be the region satisfying identity constraints.

Theorem 5.2 (Constraint Stability). If the goal constraints G define a Lyapunov function $V(s)$ that decreases along system trajectories within the region $\mathcal{R}$, then the system is **asymptotically stable** within the constraint-admissible region $\mathcal{R}$.

Proof.

By assumption, $V: \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$ is a Lyapunov function on $\mathcal{R}$ such that for all admissible trajectories $s(t) \subseteq \mathcal{R}$,

$$\dot{V}(s(t)) < 0 \text{ whenever } s(t) \neq s^*,$$

where s^* is the equilibrium state.

Restricting the system dynamics to the invariant region $\mathcal{R}$, standard Lyapunov theory implies convergence of trajectories to s^* . Constraint daemons D ensure that boundary conditions defining $\mathcal{R}$ are continuously enforced, preventing constraint violations during evolution.

Therefore, the equilibrium s^* is asymptotically stable within the constraint-admissible region $\mathcal{R}$.

5.3 Information-Theoretic Perspective

Definition 5.3 (Intelligence as Constrained Optimization). The intelligence of a COH can be quantified as:

$$\mathcal{J}(\mathcal{O}) = \max_{\pi} \mathbb{E} \left[\sum_t \gamma^t r_t \right] - \lambda \cdot \text{KL}(p_{\pi} \parallel p_{\text{constraints}})$$

where p_{π} is the policy distribution and $p_{\text{constraints}}$ encodes constraint-satisfaction probabilities.

Theorem 5.3 (Free Energy Principle Correspondence). The COH optimization objective is equivalent to minimizing variational free energy under the interpretation of constraints as priors in a Bayesian model.

6. Learning and Adaptation Theory

6.1 Neural Components as Universal Approximators

Theorem 6.1 (Universal Approximation). For any measurable function

$$h: \mathcal{X} \rightarrow \mathcal{Y}$$

and any $\epsilon > 0$, there exists a neural component

$$f_\theta \in \mathcal{N}$$

such that:

$$\|f_\theta(x) - h(x)\| < \epsilon \text{ almost everywhere}$$

provided the neural architecture has sufficient capacity.

Proof: Universal approximation results (Cybenko, 1989; Hornik et al., 1989; Yarotsky, 2017; Yun et al., 2020) ensure that neural components in COH can approximate arbitrary measurable functions.

6.2 Constrained Reinforcement Learning

Definition 6.1 (Constrained Policy Optimization). The learning problem for a COH is:

$$\max \pi_\theta \mathbb{E}_{\tau \sim \pi_\theta} [G(\tau)] \text{ subject to } \mathbb{E}_{\tau \sim \pi_\theta} [C_i(\tau)] \leq 0, \forall i$$

where C_i are constraint violation measures. The constrained optimization formulation aligns with classical CMDP theory (Altman, 1999).

Theorem 6.2 (Convergence of Constrained Learning). Under appropriate conditions (convex constraint functions, Lipschitz continuous gradients), the constrained optimization admits solutions that can be found via primal-dual methods.

6.3 Hierarchical Credit Assignment

Definition 6.2 (Hierarchical Value Decomposition). The value function for a COH decomposes as:

$$V(\mathcal{O}) = \sum_{\mathcal{O}' \in \mathcal{C}(\mathcal{O})} w_{\mathcal{O}'} V(\mathcal{O}') + V_{\text{coordination}}(\mathcal{O})$$

where $V_{\text{coordination}}$ captures emergent properties from component interactions. The hierarchical value decomposition is consistent with multi-level RL formulations (Pateria et al., 2022).

Theorem 6.3 (Optimal Decomposition). There exists an optimal weight assignment

$$\{w_{\mathcal{O}'}\}$$

that satisfies **Bellman optimality** while respecting **constraint hierarchies**.

7. Soundness and Completeness Proofs

7.1 Soundness: Internal Consistency

Definition 7.1 (Soundness). A COH theory is sound if:

1. All derivable statements are true in all models
2. The constraint system is satisfiable
3. The hierarchy is well-founded (no circular dependencies)

Theorem 7.1 (Constraint Soundness). If all identity constraints I are logically consistent (i.e., their conjunction is satisfiable), and the trigger constraints T preserve this consistency, then the COH is sound.

Proof: By induction on the **derivation length**.

- **Base case.** The initial states satisfy the identity constraints I by definition.
- **Inductive step.** Assume all states reachable in k steps satisfy I . Any state reached in $k + 1$ steps is obtained by applying a method. Each method application respects the trigger constraints T , which preserve the consistency of I . Hence the resulting state also satisfies I , which by design preserves constraint satisfaction.

Theorem 7.2 (Hierarchical Soundness). The component DAG is well-founded (contains no cycles) iff the COH has a well-defined semantics.

Proof: Acyclicity ensures that component evaluation terminates, as each component can only depend on lower-level components. This is provable by topological ordering of the DAG.

7.2 Completeness: Expressive Power

Definition 7.2 (Completeness). A COH theory is complete if it can represent:

1. Any computable function (Turing completeness)
2. Any learning system with bounded rationality
3. Any constraint-satisfaction system with finite resources

Theorem 7.3 (Computational Completeness). The COH framework is Turing-complete.

Proof sketch: We can encode a **Universal Turing Machine (UTM)** within a COH as follows:

- **$C(\text{Hierarchy})$:**
Tape cells are represented as sub-objects arranged in a hierarchical (or linked) structure.
- **$A(\text{Attributes})$:**
The machine's internal state and the head position are stored as attributes.

- ***M*(Methods):**
The transition function of the Turing Machine is implemented as methods that update tape symbols, head position, and state.
- ***I*(Identity Constraints):**
Constraints enforce tape consistency (e.g., exactly one symbol per cell, exactly one head position).
- ***T*(Trigger Constraints):**
Step-execution triggers advance the machine by one transition when the appropriate conditions are met.
- ***N*(Neural Components, optional):**
Learning components may be included to approximate or adapt transition behavior.
- ***E*(Embedding):**
The embedding maps the complete machine configuration (tape, head position, and state) into a unified latent representation.

Theorem 7.4 (Representational Completeness). For any intelligent system S with a finite description, there exists a COH $\mathcal{O}$ that simulates S .

Proof: By structural induction on the components of S . Base components map to atomic COH objects. Composite systems map to hierarchical compositions. Learning components map to neural components with appropriate architectures.

Theorem 7.5 (Constraint Expressiveness). Any constraint system expressible in first-order logic with transitive closure can be encoded in COH constraints.

Proof: Identity constraints I capture universal quantifiers. Trigger constraints T capture implication. Hierarchical composition enables transitive relationships through component nesting.

7.3 Functional Minimality and Non-Redundancy

Definition 7.3 (Functional Minimality). A COH 9-tuple is *functionally minimal* if removing any component reduces the class of systems that can be faithfully represented, in the sense that there exists at least one intelligent system whose essential features cannot be captured without that component.

Theorem 7.6 (Functional Non-Redundancy). The 9-tuple is functionally non-redundant: for each component, there exists a class of intelligent systems that cannot be adequately represented without it.

Proof sketch (per component):

- **Without C (Hierarchy):** Cannot represent hierarchical systems with nested composition.
- **Without A (Attributes):** Cannot maintain stateful information across time steps.
- **Without M (Methods):** Cannot perform actions or cause state changes.
- **Without N (Neural Components):** Cannot learn adaptively from experience.
- **Without E (Embedding):** Cannot provide unified semantic grounding across modalities.
- **Without I (Identity Constraints):** Cannot enforce invariant constraints (e.g., physical laws).

- **Without T (Trigger Constraints):** Cannot respond to events with condition–action rules.
- **Without G (Goal Constraints):** Cannot optimize behavior toward objectives.
- **Without D (Constraint Daemons):** Cannot monitor or enforce constraints in real time

For each case, a concrete counterexample system is constructed in Supplementary Appendix C.

Theorem 7.7 (Non-Redundancy of Components). The nine components are non-redundant: no component can be expressed purely in terms of the others while preserving the same representational capacity.

Proof sketch: For each component X , we exhibit two COH instances that differ only in X yet produce distinct behaviors, thereby establishing that X carries independent information. Complete independence proofs are provided in Supplementary Appendix D.

8. Applications Across Six Domains

Scope note: The following applications illustrate representational and theoretical adequacy, not empirical performance guarantees. They demonstrate that the COH formalism can capture the structure and dynamics of diverse systems, but do not constitute empirical validation or performance benchmarking.

8.1 Introduction

A central claim of Constrained Object Hierarchies (COH) is that a single, mathematically grounded framework can model intelligent behavior across widely different domains. To demonstrate this generality, we apply the COH 9-tuple formalism to six complex systems: quantum computing, systems biology, smart-city infrastructure, aerospace mission planning, financial markets, and cultural heritage preservation. These domains span physical, biological, sociotechnical, and cultural systems, each with distinct constraints, dynamics, and optimization objectives. Despite their differences, all six applications instantiate the same COH structure, showing that COH provides a unified, domain-agnostic foundation for modeling intelligent systems. Before presenting these applications, we discuss the implementability and simulatability of COH formalizations.

8.1.1 Implementability of COH

Theorem 8.1.1 (COH Constructive Implementability). For any well-formed COH specification

$$\mathcal{O} = (C, A, M, N, E, I, T, G, D),$$

there exists a **constructive implementation** $\mathcal{I}(\mathcal{O})$ in a universal computing model (e.g., a Turing machine, the lambda calculus, or a physical computer) with the following properties:

1. **Semantic Preservation.**

For every admissible state transition

$$s \rightarrow s'$$

in $\mathcal{O}$, there exists a corresponding computational step in $\mathcal{J}(\mathcal{O})$ that produces an equivalent result.

2. **Constraint Enforcement.**

All identity constraints I are maintained invariantly; trigger constraints T are evaluated before every relevant transition; and constraint daemons D execute as continuous processes.

3. **Resource Bounds:** The implementation requires resources polynomial in the specification size:

$$\begin{aligned} \text{Time}(\mathcal{J}) &= O(|\mathcal{C}|^k \cdot \text{poly}(|I|, |T|, |M|)) \\ \text{Space}(\mathcal{J}) &= O\left(\sum_{i=1}^n |\mathcal{V}_i| + |\Theta| + \text{polylog}(|\mathcal{H}|)\right) \end{aligned}$$

where k depends on the DAG depth of $\mathcal{C}$, $|\mathcal{V}_i|$ are the attribute domain sizes, $|\Theta|$ is the neural parameter size, and $|\mathcal{H}|$ is the embedding space dimension.

4. **Approximation Quality:** For neural components N , the implementation achieves an approximation error bounded by **universal approximation theorems**, with quantization effects bounded by

$$\epsilon_{\text{quant}} = O\left(\frac{1}{\text{bits}}\right).$$

Proof sketch: The proof takes six steps involving constructive implementation schema, semantic preservation, constraint enforcement, resource bounds analysis, approximation quality, and computational universality. Details are provided in Supplementary Appendix B.

8.1.2 Simulability of COH

The theoretical breadth of the COH framework, illustrated through its application across six distinct domains, must ultimately be supported by empirical validation to demonstrate practical utility. Because COH is proposed as a universal framework for world modeling and general intelligence, exhaustive real-world validation across all possible domains is neither feasible nor conceptually required. Simulation therefore becomes the primary methodological bridge between theory and practice, showing that COH models function as computational systems rather than purely abstract constructs. The Simulatability Theorem guarantees that every COH model admits a faithful computational realization, establishing simulation as a principled and sufficient validation pathway.

Theorem 8.1.2 (Simulatability). Any COH model can be simulated to arbitrary precision given sufficient computational resources.

Proof: Follows from the constructive nature of COH definitions and the computability of each component.

8.2 Quantum Computing Control Systems

8.2.1 Domain Overview

Quantum computing requires precise manipulation of quantum states under strict physical constraints such as unitarity, decoherence, and error-correction thresholds. The domain combines quantum mechanics, classical control theory, and real-time optimization—making it an ideal testbed for COH.

8.2.2 COH Formalization

$\text{COH}_{\{\text{quantum}\}} = (\mathbf{C}, \mathbf{A}, \mathbf{M}, \mathbf{N}, \mathbf{E}, \mathbf{I}, \mathbf{T}, \mathbf{G}, \mathbf{D})$

C — Components: QubitArray, ControlElectronics, ErrorCorrectionUnit, Compiler

A — Attributes: qubit_states $\in \mathbb{C}^{2^n}$, decoherence_rates $\in \mathbb{R}^n$, gate_fidelities $\in [0, 1]$

M — Methods: apply_gate(gate_matrix), measure_qubit(index), initialize_state(state)

N — Neural Components: ErrorPredictionNN, DecoherenceCompensator, OptimalControlRL

E — Embedding: QuantumStateEmbedder: $\mathcal{H} \rightarrow \mathbb{R}^{512}$ (maps high-dimensional Hilbert-space quantum states to a classical embedding)

I — Constraints: Entanglement limit: qubit_entanglement $\leq$ max_entanglement; Gate-error threshold: gate_errors $<$ threshold

T — Triggers: If decoherence_rate $>$ threshold $\rightarrow$ apply_dynamic_decoupling

G — Goals: Maximize algorithm_fidelity, Minimize error_correction_overhead

D — Daemons: CoherenceMonitor, ErrorSyndromeDaemon

8.2.3 Mathematical Guarantees

Theorem 8.2.1 (Quantum Correctness). For any quantum circuit $\mathcal{C}$ with n gates, the COH-controlled quantum system satisfies:

$$\text{fidelity}(Q(\mathcal{C}), C_{\text{ideal}}) \geq 1 - n\varepsilon_{\text{gate}} - \varepsilon_{\text{meas}} - \varepsilon_{\text{decoh}},$$

where each error term is bounded by identity constraints and regulated by daemons.

8.3 Biological System Simulation

8.3.1 Domain Overview

Systems biology involves multi-scale modeling of molecular networks, cells, tissues, and organs. These systems exhibit nonlinear dynamics, stochasticity, and complex regulatory constraints.

8.3.2 COH Formalization

$\text{COH}_{\{\text{bio}\}} = (\mathbf{C}, \mathbf{A}, \mathbf{M}, \mathbf{N}, \mathbf{E}, \mathbf{I}, \mathbf{T}, \mathbf{G}, \mathbf{D})$

C — Components: MolecularNetwork, CellPopulation, TissueStructure, OrganSystem

A — Attributes: gene_expression $\in \mathbb{R}^m$, metabolite_concentrations $\in \mathbb{R}^n$, cell_states $\in \text{Enum}^k$

M — Methods: simulate_reaction_network(t), apply_drug_concentration(drug, c), mutate_gene(gene)

N — Neural Components: GeneRegulationPredictor, MetabolicFluxEstimator, CellFateClassifier

E — Embedding: BiologicalStateEncoder: multi-omic data $\rightarrow \mathbb{R}^{1024}$ unified biological state representation

I — Constraints: Mass_balance: production = consumption + accumulation; Energy_conservation: ATP_production $\geq$ cellular_maintenance

T — Triggers: If nutrient_depletion $\rightarrow$ activate_starvation_response; If DNA_damage_detected $\rightarrow$ activate_repair_pathways

G — Goals: Maximize model_accuracy(experimental_data), Minimize computational_cost

D — Daemons: HomeostasisDaemon, EnergyBalanceMonitor, ToxicityPredictor

8.3.3 Mathematical Guarantees

Theorem 8.3.1 (Biological Plausibility). For any dataset D , the COH biological model satisfies:

$$KL(M_{\text{pred}} \parallel D_{\text{obs}}) \leq \varepsilon_{\text{exp}} + \varepsilon_{\text{model}} + \varepsilon_{\text{stoch}},$$

with each term bounded by identity constraints.

8.4 Smart-City Infrastructure Management

8.4.1 Domain Overview

Smart cities integrate energy, transportation, water, and communication systems. These infrastructures require real-time optimization, resilience, and equitable service distribution.

8.4.2 COH Formalization

COH_{{\text{smartcity}}} = (C, A, M, N, E, I, T, G, D)

C — Components: EnergyGrid, TransportationNetwork, WaterSystem, CommunicationInfrastructure

A — Attributes: energy_demand: TimeSeries, traffic_flow: Graph $\rightarrow \mathbb{R}$, water_levels: SpatialField

M — Methods: route_vehicles(source, dest), allocate_power(region, amount), manage_wastewater()

N — Neural Components: DemandForecaster: Transformer, TrafficOptimizer: GNN, ResourceAllocator: MARL

E — Embedding: CityStateEncoder: multi-modal urban data $\rightarrow \mathbb{R}^{2048}$ unified representation

I — Constraints: Flow conservation: Σ inflows = Σ outflows; Utility service guarantee: service_level $\geq$ SLA_minimum

T — Triggers: If traffic_congestion > threshold $\rightarrow$ activate_alternate_routes; If power_outage_detected $\rightarrow$ reroute_energy_supply

G — Goals: Minimize carbon_emissions, Maximize resource_utilization_efficiency, Maximize citizen_quality_of_life

D — Daemons: ResilienceMonitor, SustainabilityDaemon, EquityEvaluator

8.4.3 Mathematical Guarantees

Theorem 8.4.1 (Urban Efficiency). The COH-managed city achieves Pareto-optimal solutions to multi-objective optimization problems:

$$\min_{x \in X} [f_1(x), \dots, f_k(x)]^T,$$

where X is the constraint-satisfying region.

8.5 Aerospace Mission Planning

8.5.1 Domain Overview

Aerospace missions involve hybrid dynamics, strict safety constraints, and long-horizon planning under uncertainty.

8.5.2 COH Formalization

COH $\{\text{space}\}$ = (C, A, M, N, E, I, T, G, D)

C — Components: OrbitalDynamics, PropulsionSystem, PowerManagement, ScientificInstruments

A — Attributes: position $\in \mathbb{R}^3$, velocity $\in \mathbb{R}^3$, fuel_mass $\in \mathbb{R}$, instrument_health $\in [0, 1]^m$

M — Methods: execute_orbital_maneuver(Δv), acquire_science_data(target), communicate_ground()

N — Neural Components: TrajectoryOptimizer: NeuralODE, AnomalyDetector: Autoencoder, ResourcePredictor: LSTM

E — Embedding: SpacecraftStateEncoder: combines orbital elements, resource states, and mission objectives

I — Constraints: Orbital energy conservation: $E = \frac{v^2}{2} - \frac{\mu}{r}$; Mass balance: $m_{\text{final}} = m_{\text{initial}} - \int \dot{m} dt$

T — Triggers: If radiation_level > safe_limit $\rightarrow$ shield_instruments; If communication_loss > duration $\rightarrow$ enter_autonomous_mode

G — Goals: Maximize scientific_return, Minimize fuel_consumption, Ensure mission_success_probability > 0.99

D — Daemons: RadiationMonitor, SystemHealthDaemon, ContingencyPlanner

8.5.3 Mathematical Guarantees

Theorem 8.5.1 (Mission Safety). The probability of mission failure satisfies:

$$P(\text{failure}) \leq \sum_i \lambda_i T + \sum_j \int_0^T h_j(t) dt,$$

with hazard rates bounded by constraints and mitigated by daemons.

8.6 Financial Market Ecosystem Simulation

8.6.1 Domain Overview

Financial markets involve strategic agents, regulatory constraints, and systemic risk propagation.

8.6.2 COH Formalization

COH\$_{\text{finance}}\$ = (C, A, M, N, E, I, T, G, D)

C — Components: InvestorPopulation, MarketMakerNetwork, ClearingHouse, RegulatoryBody

A — Attributes: asset_prices $\in \mathbb{R}^m$, portfolio_holdings $\in \mathbb{R}^{n \times m}$, market_microstructure = OrderBook

M — Methods: submit_order(asset, quantity, type), clear_market(), adjust_regulations(params)

N — Neural Components: PricePredictor: Transformer, AgentBehaviorModel: IRL, SystemicRiskDetector: GAN

E — Embedding: MarketStateEncoder: combines prices, volumes, sentiments, and network structures

I — Constraints: No_arbitrage: $\text{expected_return} \geq \text{risk_free_rate} + \beta \cdot \text{market_premium}$; Market_clearing: $\Sigma \text{buy_orders} = \Sigma \text{sell_orders}$

T — Triggers: If volatility > VIX_threshold $\rightarrow$ activate_circuit_breakers; If liquidity < minimum $\rightarrow$ inject_market_making

G — Goals: Maximize market efficiency: $1 - |\text{fundamental_value} - \text{market_price}| / \text{fundamental_value}$; Minimize systemic risk

D — Daemons: FlashCrashDetector, ManipulationMonitor, LiquidityDaemon

8.6.3 Mathematical Guarantees

Theorem 8.6.1 (Market Stability). The COH-simulated market achieves efficient price discovery:

$$\text{Efficiency} = 1 - O\left(\frac{1}{\sqrt{N_{\text{informed}}}}\right),$$

consistent with empirical findings on the square-root law (Tóth et al., 2016; Benzaquen & Bouchaud, 2018; Bucci et al., 2019).

8.7 Cultural Heritage Preservation

8.7.1 Domain Overview

Cultural heritage preservation requires balancing conservation, access, authenticity, and long-term risk management.

8.7.2 COH Formalization

COH\$_{\text{cultural}}\$ = (C, A, M, N, E, I, T, G, D)

C — Components: ArtifactScanner, MaterialAnalyzer, HistoricalContextDB, ConservationPlanner

A — Attributes: artifact_state $\in \text{ImageSpace} \times \text{MaterialSpace} \times \text{StructuralSpace}$; material_composition $\in \mathbb{R}^{\text{elements}}$; deterioration_rate $\lambda \in \mathbb{R}^+$ (Arrhenius form: $\lambda = A \cdot \exp(-E_a/(R \cdot T))$); historical_significance $\sigma \in [0, 1]$

M — Methods: digitize_artifact(resolution), analyze_materials(technique), recommend_conservation(treatment), control_environment(parameter, value)

N — Neural Components: StyleRecognizer (CNN), DeteriorationPredictor (PINN), SemanticInterpreter (VLM), AuthenticityVerifier (GAN)

E — Embedding: CulturalArtifactEncoder: fuses visual, material, historical, and contextual information

I — Constraints: Conservation ethics: Minimal_intervention $\wedge$ Reversibility $\wedge$ Documentation; Authenticity: Original_material $\geq$ threshold; Environmental: RH 50% $\pm$ 5%, T 20°C $\pm$ 2°C, light $\leq$ 50 lux

T — Triggers: If humidity > safe_range $\rightarrow$ activate_climate_control; If deterioration detected $\rightarrow$ schedule_conservation; If handling requested for fragile artifact $\rightarrow$ require special protocol

G — Goals: Maximize information_preservation = $-\Delta H(\text{artifact_state})$; Maximize accessibility; Minimize degradation_risk; Maximize cultural_value

D — Daemons: EnvironmentalMonitor, AuthenticityVerifier, AccessControlDaemon, DeteriorationTracker

8.7.3 Mathematical Guarantees

Theorem 8.7.1 (Preservation Guarantee). For artifact A ,

$$I(T) \geq I_0 \exp \left(- \int_0^T \lambda(t) dt \right),$$

where $\lambda(t)$ is the deterioration rate mitigated by conservation interventions.

8.8 Cross-Domain Analysis

8.8.1 Shared Mathematical Structure

Across all six domains, the same COH 9-tuple structure appears, demonstrating:

- universal hierarchical decomposition
- constraint satisfaction as a unifying principle
- consistent treatment of physical, biological, social, and cultural constraints
- modularity and compositionality across scales

8.8.2 Flexibility Across Dimensions

COH supports:

- **scale adaptability** (from qubits to megacities)
- **temporal flexibility** (from femtoseconds to centuries)

- **uncertainty modeling** (quantum, stochastic, economic, environmental)
- **multi-objective optimization** (efficiency, safety, equity, preservation)

8.8.3 Unified Simulatability

The simulatability theorem (Theorem 8.1.2) holds across all domains, confirming that COH models are both theoretically expressive and practically implementable.

9. Connections to Existing Theories

9.1 Relationship to Free Energy Principle

Theorem 9.1 (COH as Generalized Free Energy Minimization). The COH optimization objective generalizes the free energy principle:

$$F(s) = \mathbb{E}_q[\log q(\psi | s) - \log p(\psi, s | \theta)] + \text{ConstraintViolation}(s)$$

where constraints act as additional priors.

9.2 Relationship to Hierarchical Reinforcement Learning

COH provides a formal foundation for hierarchical RL, with:

- Options corresponding to method sequences
- Subgoals corresponding to goal constraints
- Termination conditions corresponding to trigger constraints

9.3 Relationship to Cognitive Architectures

COH unifies aspects of:

- ACT-R (production rules as trigger constraints)
- SOAR (goal hierarchies as constraint hierarchies)
- CLARION (dual-process as neural vs. symbolic components)

10. Implications for AGI Implementation

10.1 World Modeling as Constrained Hierarchical Projection

Theorem 10.1 (World Model Fidelity). Let W be a real-world system and $\mathcal{O}_W$ its COH representation. For any finite observation sequence $o_{1:T}$ from W , there exists a COH model $\mathcal{O}_M$ such that the Kullback–Leibler divergence between the predicted and observed state distributions is bounded by the expressiveness of the constraint system:

$$D_{KL}(P_W(s_t | o_{1:t}) \parallel P_M(s_t | o_{1:t})) \leq \alpha_H H(W) + \beta_C C(I) + \gamma_N \dim(N)$$

where $H(W)$ is the hierarchical complexity of W , $C(I)$ measures the completeness of identity constraints, and $\dim(N)$ represents the capacity of neural components. The constants α_H , β_C , and γ_N are domain-dependent.

Proof sketch: Hierarchical decomposition $\$C\$$ ensures structural alignment between model and world. Identity constraints $\$I\$$ restrict deviations from physical laws. Neural components $\$N\$$ provide universal approximation for residual dynamics. The bound follows by chaining approximation errors across hierarchical levels, with daemons $\$D\$$ reducing residual divergence through constraint enforcement.

10.2 Jagged Intelligence Avoidance

Theorem 10.2 (Smooth Capability Generalization). For a COH system $\mathcal{O}$ trained on tasks $\mathcal{T} = \{T_1, \dots, T_m\}$, the performance gap on a new task T_{m+1} satisfies :

$$|R(T_{m+1}) - \frac{1}{m} \sum_{i=1}^m R(T_i)| \leq L \cdot d_C(S_{m+1}, \text{span}\{S_1, \dots, S_m\})$$

where d_C is the distance in constraint space, L is the Lipschitz constant of the policy with respect to constraints, and S_i is the constraint-admissible region for task T_i . The bound tightens as the constraint system becomes more complete and orthogonal.

Proof sketch: **Jagged intelligence** arises when systems perform well on training distributions but fail under distribution shift. COH reduces this risk through identity constraints I that enforce invariants, hierarchical composition C that enables systematic recombination of capabilities, and daemons D that detect constraint violations indicative of distribution shift. The metric d_C quantifies alignment between task requirements, with smaller distances yielding smoother generalization.

10.3 Agentic System Implementation

Theorem 10.3 (Agent Construction from COH). Any autonomous agent specification

$$A = (\text{Percepts}, \text{Actions}, \text{Goals}, \text{Ethics})$$

- can be implemented as a COH $\mathcal{O}_A$ with the following properties:

- **Perceptual grounding:**

Emaps raw percepts to semantic embeddings consistent with I .

- **Action realizability:**

Each $a \in \text{Actions}$ corresponds to a method sequence

$$m_{i_1} \circ \dots \circ m_{i_k} \in M^*$$

respecting the trigger constraints T .

- **Goal achievement:**

System behavior converges to states maximizing G , subject to constraints I and T .

- **Ethical compliance:**

Ethical principles are encoded as prioritized constraints

$$(I_{\text{ethics}}, T_{\text{ethics}})$$

with **lexical priority** over non-ethical constraints.

The agent's decision complexity grows as

$$O(\log |C| \cdot \text{poly}(|I|, |T|)),$$

rather than exponentially in state-space size.

Proof sketch: The architecture arises directly from the COH components: perception is realized as attribute updating via A ; world modeling is achieved through hierarchical decomposition C combined with neural prediction N ; planning is formulated as constraint-satisfying method selection guided by goals G ; and ethical behavior is ensured through prioritized constraints enforced by I , T , and D . Hierarchical structure yields divide-and-conquer reasoning, while constraints prune the search space. Lexicographic ordering ensures ethical precedence (e.g., safety before efficiency).

10.4 Key Benefits of COH for AGI

Engineering Principles

- Modularity via hierarchical composition
- Safety through constraint enforcement
- Adaptability through neural components
- Interpretability through explicit constraint systems

Ethical Considerations

The constraint framework naturally encodes ethical principles as:

- Identity constraints for fundamental rights
- Trigger constraints for deontological rules
- Goal constraints for consequentialist optimization

Scalability Properties

Theorem 10.4 (Compositional Scalability). The complexity of reasoning in a COH grows polynomially with the number of components, not exponentially.

Proof sketch: Hierarchical structure enables divide-and-conquer reasoning, while constraints limit cross-component interactions, preventing combinatorial explosion.

11. Conclusion

11.1 General Summary

This paper has presented a unified and mathematically rigorous foundation for **Constrained Object Hierarchies (COH)** as a theory of general intelligence. By formalizing intelligence as constrained hierarchical optimization, COH integrates structural composition, learning, prediction, and control within a single framework. Through its category-theoretic formulation, dynamical-systems interpretation, learning theory, and constraint-based semantics, COH establishes a coherent foundation for modeling adaptive, interpretable, and domain-general intelligent systems.

Beyond its theoretical development, COH demonstrates substantial **cross-domain generality**, capturing the structure and dynamics of systems as varied as quantum control, systems biology, smart-city infrastructure, aerospace missions, financial markets, and cultural heritage preservation. This breadth illustrates the expressive power of the framework and supports its potential as a unifying model for diverse forms of natural and artificial intelligence.

11.2 Future Directions

Several directions for further work emerge naturally from the present theory:

1. **Computational complexity and tractability** of inference, optimization, and hierarchical reasoning within COH-structured systems.
2. **Learning and adaptation of constraints**, including methods for discovering or refining identity, trigger, and goal constraints from data.
3. **Higher-order categorical generalizations**, extending COH to enriched categories, operadic structures, or multi-level semantic morphisms.
4. **Quantum extensions**, exploring how COH can incorporate quantum machine-learning components and hybrid classical–quantum intelligent architectures.

11.3 Final Remarks

COH provides a robust, neuroscience-grounded, and mathematically principled framework for understanding and engineering general intelligence. Its integration of hierarchical composition, constraint-driven regulation, predictive optimization, and adaptive learning positions it uniquely among existing theories of intelligence. By offering a unified account that spans symbolic, neural, dynamical, and probabilistic perspectives, COH lays a promising foundation for future AGI research, with implications for safety, interpretability, and scalable autonomous behavior.

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Supplementary materials (available upon request):

- **Appendix A:** Notation table and guidelines for translating COH formalizations into executable models
- **Appendix B:** Complete proofs of all theorems
- **Appendix C:** Counterexample constructions for functional non-redundancy
- **Appendix D:** Independence proofs for component non-redundancy